Elie Attal

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Education

2024 – present **PhD in applied mathematics**
CMAP, École Polytechnique
Advisors: E. Abi Jaber and M. Rosenbaum.
Thesis: Interacting agents, jump processes and Volterra processes.

2023 – 2024 **M2 Probability and Finance**
Sorbonne Université / École Polytechnique

2020 – 2023 **Ingénieur polytechnicien master's degree**
École Polytechnique
Master's degree in applied mathematics.

Professional experience

4 months, 2026 **Quantitative research intern**
Bloomberg L.P., New York
Developed numerical methods for path-dependent option pricing under rough volatility models. Carried out as part of the Bloomberg Quantitative Finance PhD Fellowship.

6 months, 2024 **Quantitative research intern**
Qube Research & Technologies, Paris
Studied the limiting eigenvector overlaps between large random matrices and their submatrices. Published as two papers in *Journal of Physics A: Mathematical and Theoretical*.

5 months, 2023 **Quantitative research intern**
DRW Holdings, London
Designed a quoting model for RFQ platforms combining machine learning and probability.

3 months, 2022 **Quantitative research intern**
ITO33, Paris
Extended the Black–Scholes framework to price convertible bonds under takeover risk. Implemented the model in C++ in the company's pricing software.

Awards

2026 **QRT PhD Prize**
For the talk *Affine structure of the Brownian signature* at the London–Paris Bachelier Workshop on Mathematical Finance.

2025 **Bloomberg Quantitative Finance PhD Fellowship**

Publications

- **Simulating integrated Volterra square-root processes and Volterra Heston models via inverse Gaussian**
with E. Abi Jaber. *Finance and Stochastics*, to appear, 2026. arXiv:2504.19885

- **From hyper roughness to jumps as $H \to -1/2$**
with E. Abi Jaber and M. Rosenbaum. *The Annals of Applied Probability* 36(4), 3635–3660, 2026. arXiv:2503.16985

- **Eigenvector overlaps of random covariance matrices and their submatrices**
  with R. Allez. *Journal of Physics A: Mathematical and Theoretical* 58, 205003, 2025. arXiv:2501.08768
- **Interlacing eigenvectors of large Gaussian matrices**
  with R. Allez. *Journal of Physics A: Mathematical and Theoretical* 58, 045003, 2025. arXiv:2409.17086

Preprints

- **Stochastic control with signatures via Riccati equations on the tensor algebra**
  with E. Abi Jaber and D. Sotnikov. 2026. arXiv:2607.03986
- **Affine structure of the Brownian signature**
  with E. Abi Jaber and D. Sotnikov. 2026. arXiv:2606.29622
- **Volterra clocks and their pure-jump limits: hitting times of curved boundaries**
  with E. Abi Jaber and A. Søjmark. 2026. arXiv:2605.29073
- **Efficient simulation of Hawkes processes using their affine Volterra structure**
  with E. Abi Jaber and D. Sotnikov. 2025. arXiv:2511.13554

Talks

Conferences and workshops

Sept. 2026	London–Paris Bachelier Workshop on Mathematical Finance, Paris *(invited)* *Affine structure of the Brownian signature — QRT PhD Prize*
Sept. 2026	International Conference on Computational Finance, Oxford *Simulating integrated Volterra square-root processes and Volterra Heston models via inverse Gaussian*
Apr. 2026	Stochastic Control and Games for Risk & Regulation, Hammamet *Simulating affine Volterra processes via inverse Gaussian*
Jan. 2026	18th Bachelier Colloquium on Financial Mathematics and Stochastic Calculus, Métabief *From hyper-roughness to jumps as $H \to -1/2$: the affine case and beyond*
Sept. 2025	Stochastics & Computational Finance, Lisbon *Simulating affine Volterra processes via inverse Gaussian*
July 2025	Vienna Congress on Mathematical Finance, Vienna *From hyper roughness to jumps as H goes to $-1/2$*

Seminars

Oct. 2026	Seminar of the CMAP MathsFi group, École Polytechnique *Volterra clocks: going beyond the affine structure*
Jan. 2026	CMAP PhD Seminar, École Polytechnique *Introduction to non-Markovian diffusions in Mathematical Finance*
Dec. 2025	Seminar of the CMAP MathsFi group, École Polytechnique *Simulating affine Volterra processes via inverse Gaussian: the iVi schemes*
May 2025	Local Seminar, Vega Institute, Moscow (online) *From hyper roughness to jumps as $H \to -1/2$*
Mar. 2025	Bachelier Seminar (PhD session), IHP, Paris *From hyper roughness to jumps as $H \to -1/2$*
Mar. 2025	Seminar of the CMAP MathsFi group, École Polytechnique *From hyper roughness to jumps as $H \to -1/2$*

Teaching

2026 – 2027	**École Polytechnique** Supervision of M1 group projects: *Stability of eigenvectors of covariance matrices in finance*; *Heat equation and Cole–Hopf transform: entire series representations and numerical schemes*
2025 – 2026	**École Polytechnique** Markov processes and applications (M1), tutorial classes Probability (L3), tutoring **ENSAE** Supervision of an M1 group project: *Toward a new stochastic model for integrated variance: insights from SPX and VIX*
2024 – 2025	**École Polytechnique** Markov processes and applications (M1), tutorial classes Probability (L3), tutoring Supervision of an M1 internship: *Numerical schemes for Riccati–Volterra equations*

Skills

Languages	French (native), English (fluent), Spanish (conversational)
Programming	Python (numpy, scipy, pandas, scikit-learn), SQL, Git, LaTeX, Power BI